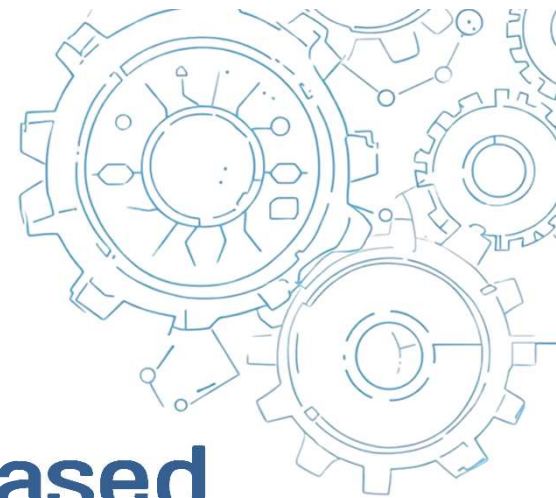




Odor source localization RL based on *Drosophila* larvae chemotaxis

25.06.08
NISL LAB MEETING

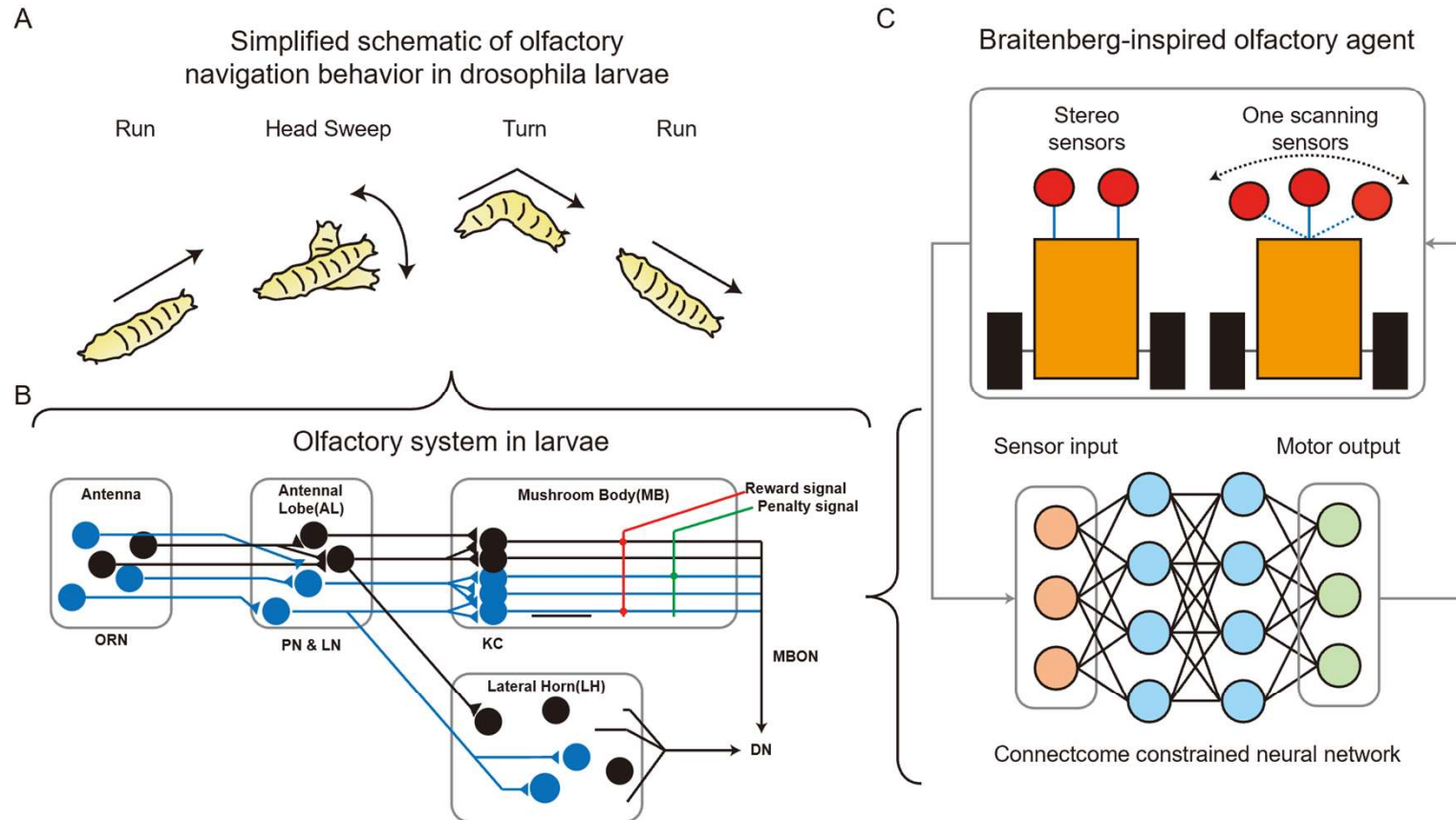
Department of Biomedical engineering

Hyunseo In



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Drosophila larvae chemotaxis



"We could apply these features to OSL by RL"
⟨Connectome & Active sensing⟩



Baseline Model

DETAILS

Bilateral odor sensing (0.15 mm gap) — left/right
Left-right difference $\times 80$ (proportional control)

$$\omega_{body} = \tanh(80G \cdot \delta(R, L)), \quad \delta(R, L) = \frac{c_L - c_R}{c_L + c_R}$$

Environments

noise-free / static-noise / dynamic-noise

Result

Steers the body directly

Success rate degrade as noise rises

Active sensing(Cast) used as noise rises

Better than any other RL agents

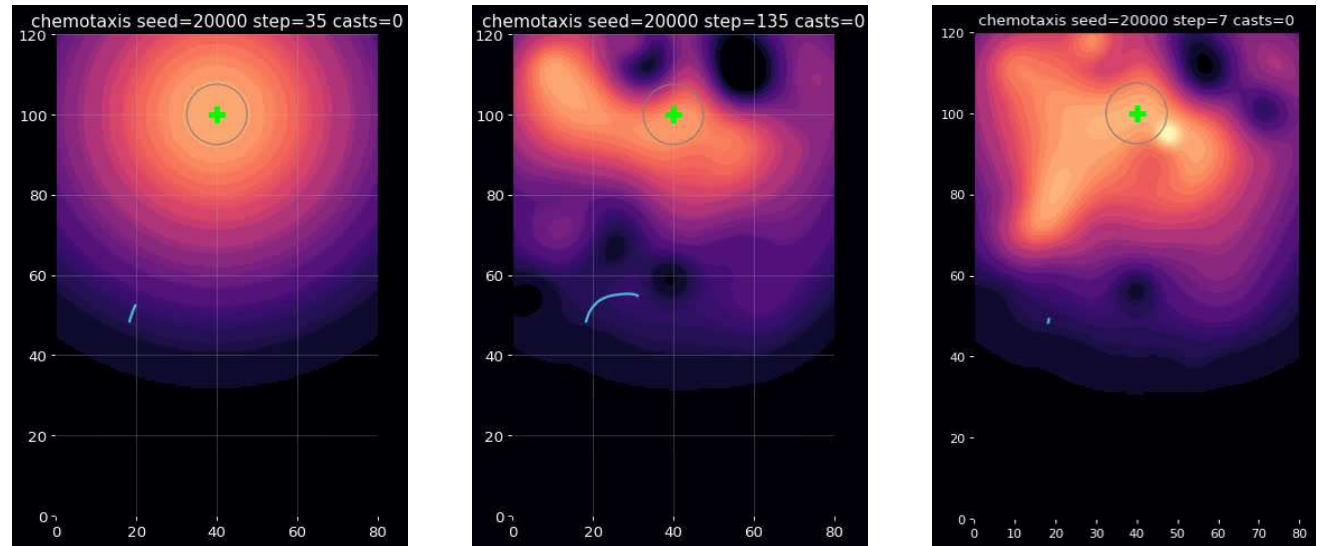


Figure2.a: baseline model agent's trajectory of each three environments ($S0_{\sigma 0}, S1_{\sigma 1.0}, S2_{\sigma 1.0}$)

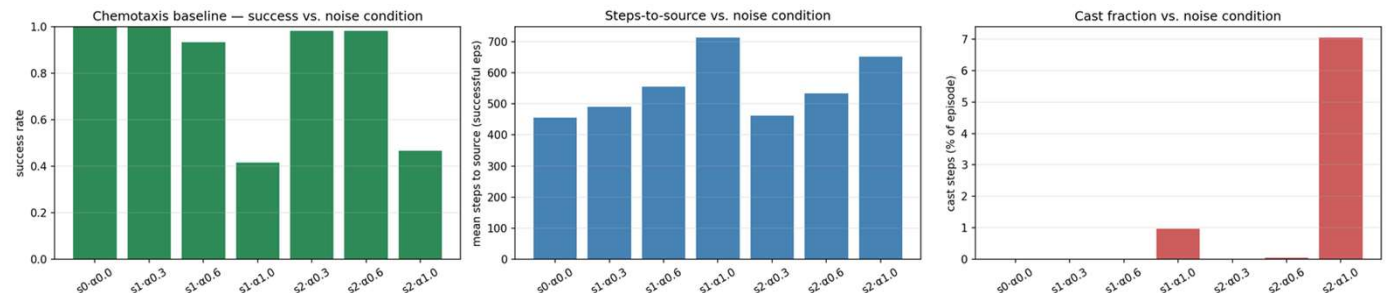


Figure2.b: baseline model agent's success rate, step-to-source, cast ratio

"Simple Optimization can solve Gaussian 2D-OSL"



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RL framework design

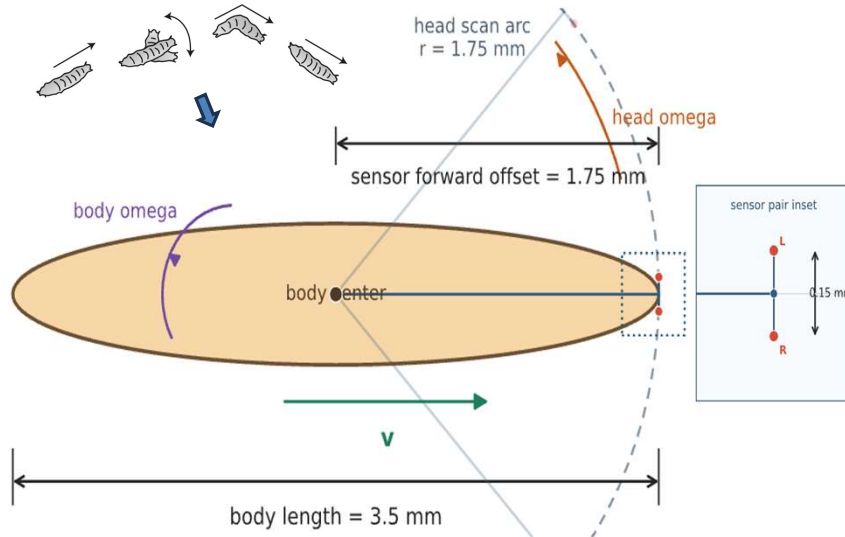


Figure3.a: Agent design, based on Drosophila larvae features

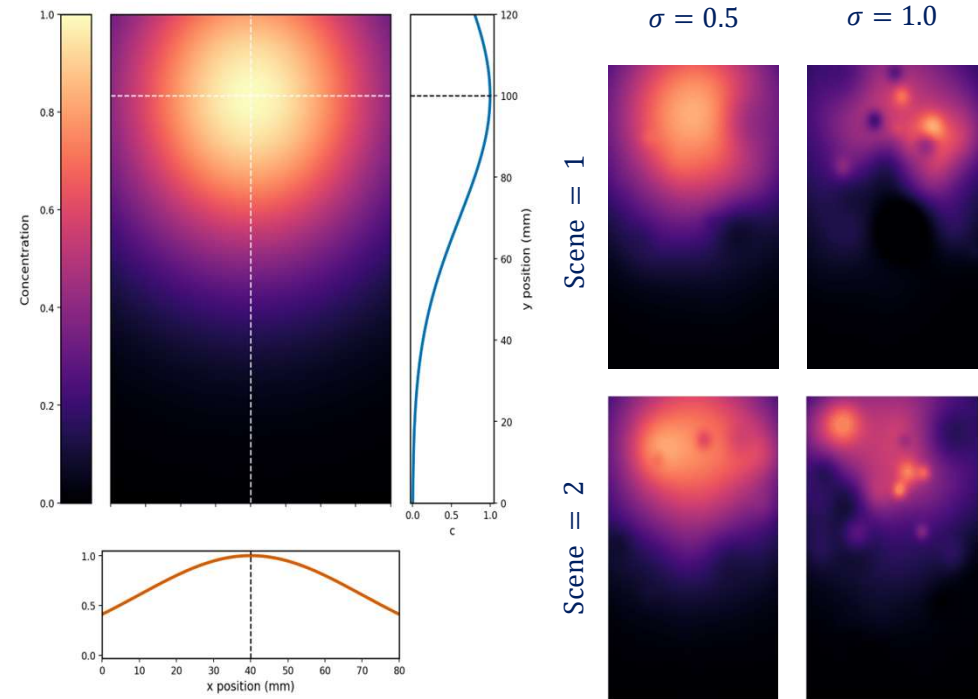


Figure3.b: Gaussian 2D environment

"Biologically designed RL agent and environments"



Policy: GRU recurrent network

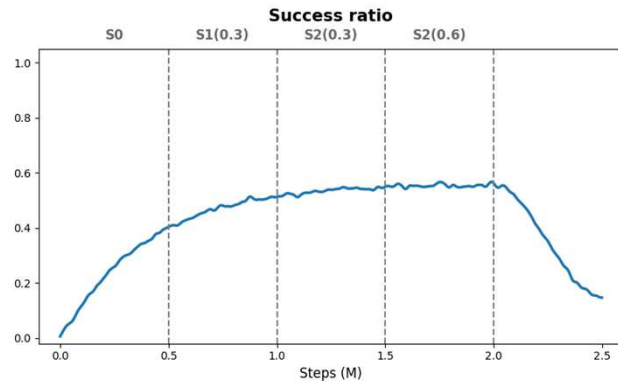


Figure4.a: PPO with GRU layer model's success ratio($S2, \sigma 0.6$) through curriculum learning

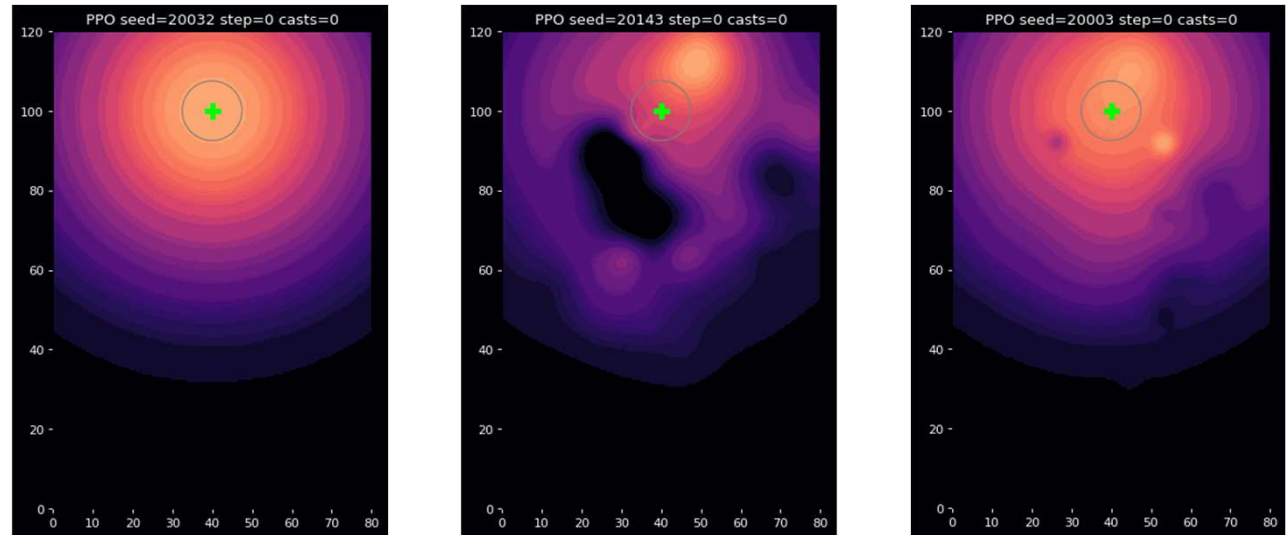


Figure4.b: GRU model agent's trajectory of each two environments ($S0_{\sigma 0}, S1_{\sigma 1.0}, S2_{\sigma 0.5}$)

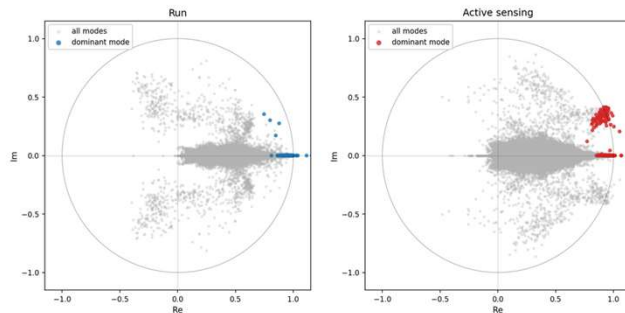


Figure4.d: Eigenvalue of Jacobian matrix (colored ones are the dominants; largest $|\lambda|$)

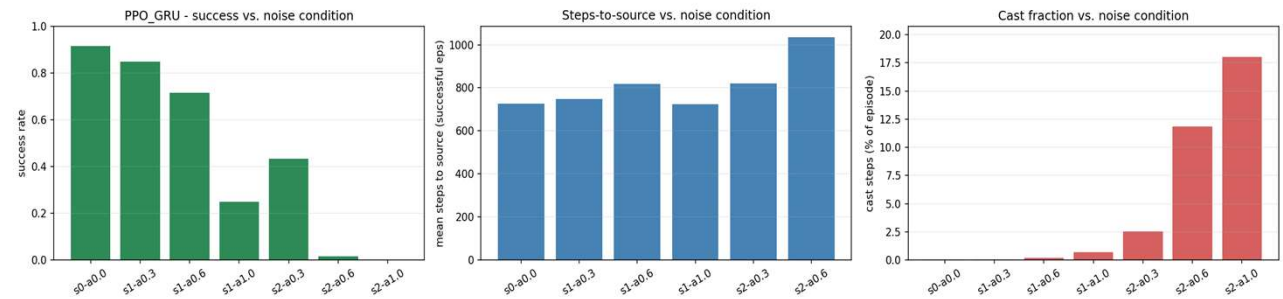


Figure4.c: baseline model agent's success rate, step-to-source, cast ratio

"The GRU-based RL agent learned active sensing."



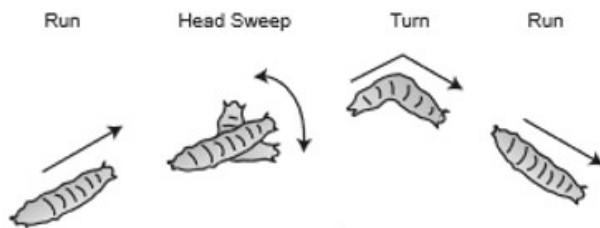
Trial and errors

Too small difference of Right Left

Bilateral odor sensing (0.15 mm gap) — left/right

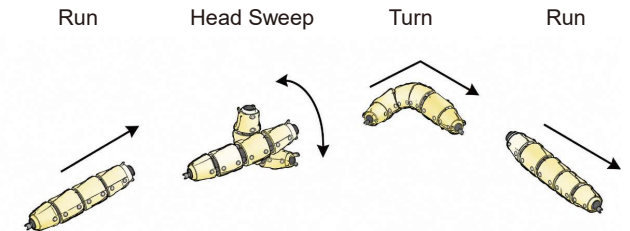
Without 80(proportional control) Gain, $R-L = 0.5\%$

Absolute values of R, L odor concentration only is too hard for simple RNN



Too small parameters

Connectome model is 17K, GRU is 540K, almost 1/30 of GRU model



Too difficult task

Connectome is basically for a simple classification,
not for E2E plume-to-steer model,
better for perception-only model

Reinforcement Learning is hard

Too many hyperparameters and too slow



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Thank You

